

Date: Jan 21, 2021

Experiment number: 1

Title: Recrystallization

Objective: The objective of this experiment was to purify or separate a contaminated benzoic acid from its impurities by the technique of recrystallization and to calculate how much percent of benzoic acid was recorded after recrystallization

Theory

- **Recrystallization** is a purification technique for solid compounds; an impure solid compound is mixed with hot solvent to form a saturated solution. As the solvent cools, the solution becomes saturated with the solute and the solute crystallizes out (reforms a solid). As the crystal develops, impurities are excluded from the crystal lattice, thereby completing the purification process. It is used to separate two compounds with very different solubility properties, and as the temperature increases, the amount of solute dissolved in the solvent increases.

Apparatus/ material used

- Erlenmeyer flask
- Measuring cylinder
- Iron stand
- Clamp
- Steam less funnel
- Fluted filter paper
- Spatula
- Suction filter(formed by assembling Buchner funnel+ suction flask)
- Hot plate
- Electronic balance

Chemicals used

- Impure Benzoic acid
- Water

Procedure

Before the start of our experiment 2gm of impure Benzoic acid was already measured and prepared by our instructor. Then the 2gm of impure benzoic acid was added to an Erlenmeyer flask. After that 100ml of water was measured and then added to the flask containing the impure benzoic acid. The flask containing the impure benzoic acid and water was heated by being placed on a hot plate while the solution was being stirred. The impurity was dissolved completely after a

few minutes. Then the solution was poured slowly into a flask with a seamless funnel on it containing a filter paper, and only the impurity was left. The flask was put inside ice to let it cool and crystalize. After a few minutes, the flask was taken out of the ice.

A new fluted filter was weighted, and by following the steps told by our instructor, the fluted filter was prepared by folding it more than once in certain directions. Then the folded fluted paper was placed on a Buchner funnel. A bit of water was added to the filter paper on the assembled suction filter to make it wet. The Buchner funnel was attached to the suction filter, a water aspirator was connected to it, and the tap water was turned on. Following that the crystalized solution was added onto the Buchner funnel containing the fluted filter paper, slowly a little water was added onto the folded fluted filter paper to make sure everything on it was filtered completely. An open paper box was prepared and our names were written on it. The filter paper containing the filtered sample was placed on the paper box. Two weeks later the sample had dried, and was ready for measurement. The mass of the sample was measured using an electronic balance.

Results and discussion

Before recrystallization

- The weight of impure benzoic acid= 2gm
- The weight of filtered paper= 1.84gm

After recrystallization:

- The weight of benzoic acid= weight of recrystallized benzoic acid- weight of filter paper = 4.09gm- 1.84gm= 2.25gm

% recovery=(final mass of pure benzoic acid÷ initial mass of crude/ impure crystal) × 100

$$= (2.25\text{gm} \div 2\text{gm}) \times 100 = 112.5\%$$

According to the process of recrystallization, the final mass should have been less than the initial mass before recrystallization, but after asking our instructor we have realized that the electronic balance has some problems, due to that the mass we ended with was higher than the initial one.

The percentage recovery is more than 100% because of the problem of the balance which resulted in a higher mass than the benzoic acid before recrystallization. If the balance and the environmental conditions were correct, then the mass should have been 2gm or less than that.

Quiz question

1. How does a water aspirator create partial vacuum?

- The water aspirator pump is essentially a pipe with a narrowing in it. As water flows through that narrowing, it speeds up and its pressure drops—it's exchanging its pressure

energy for kinetic energy. A water aspirator works when water from the tap is forced at high velocity through a small nozzle. At the smallest part of the nozzle, a partial vacuum is created. A tube attached to this portion and the vacuum draws fluid from through that tube.

2. What is the significance of the term “like dissolves like”?

- When making a solution, it's useful to know if one thing will dissolve in another. If somebody wants to make a liquid solution of one chemical, it won't be good if the liquid doesn't dissolve and sit at the bottom because we chose the wrong solvent. The best way to tell if something will dissolve is to look at the polarities of the solvent and the solute. If the polarities of the solvent and solute match, then the solute will probably dissolve. If the polarities of the solvent and solute are different (one is polar, one is nonpolar), the solute probably won't dissolve. The principle “like dissolves like” is that it tells us which substances have similar intermolecular force and tend to dissolve completely. For example polar solvent tends to dissolve polar solutes and non-polar solvent tends to dissolve non-polar solute.

3. What are the differences between an ordinary paper and filter paper?

- Filter paper is more porous and will catch the containment more efficiently than ordinary paper. It increases the speed of filtration by letting air enter the flask along its sides to bring about pressure equalization. In addition, the fluted filter paper has a larger surface area through which the solvent can seep. The folds also create space between the filter paper and glass funnel, allowing for displaced air to more easily exit the flask as liquid drains. Normal (ordinary) paper lacks the above qualities, and cannot separate complex material but filter paper can do this.

4. Look at the flame generated by the Bunsen burner. What are the different parts of the flame? Which part is hottest?

- In our experiment we used a hot plate instead of a Bunsen burner, but if we had used a Bunsen burner we could have the following types of flames. 1. Reddish flame 2. Yellow flame 3. Blue flame. Blue flame is the hottest flame from the above 3 flames.

5. Some solids make perfect crystals, while others are amorphous. What may be the cause for this?

- Almost any substance can solidify in amorphous form if the liquid phase is cooled rapidly enough. Some solids however are intrinsically amorphous, because either their components can't fit together well enough to form a stable crystalline lattice or they contain impurities that dispartate lattice. So they have a short order arrangement of molecules. Crystals often form in nature when liquids cool and start to harden. Certain molecules in the liquid gather together as they attempt to become stable. They do this in a uniform and repeating pattern that forms the crystal.



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